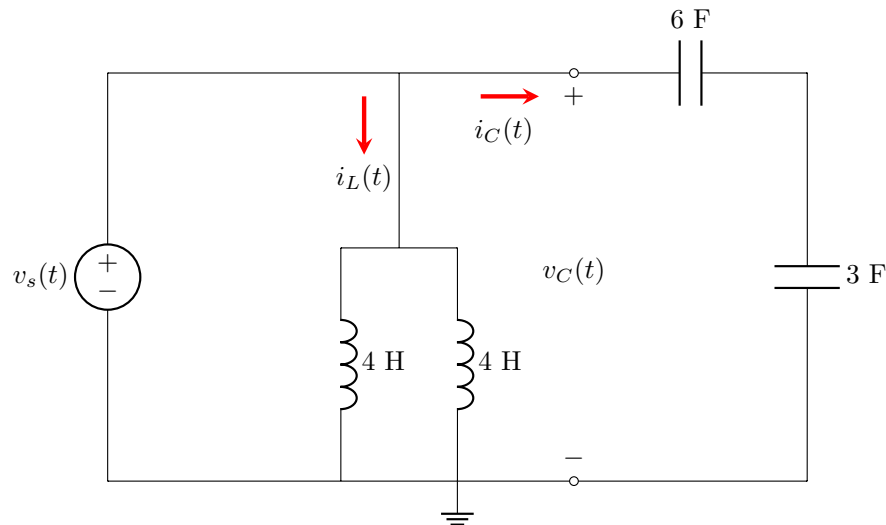


Section 3 / Afdeling 3

Questions 33 to 35 [6]

In the circuit below, $v_s(t) = \alpha e^{-0.2t}$ V and $i_L(0) = 2$ A.
The value of α is given in ClickUP.



33 Determine $v_C(0)$. [2]

34 Determine $i_C(5)$. [2]

35 Determine $i_L(5)$. [2]

33. $v_s(0) = v_C(0) \Rightarrow v_C(0) = \alpha$ (V).

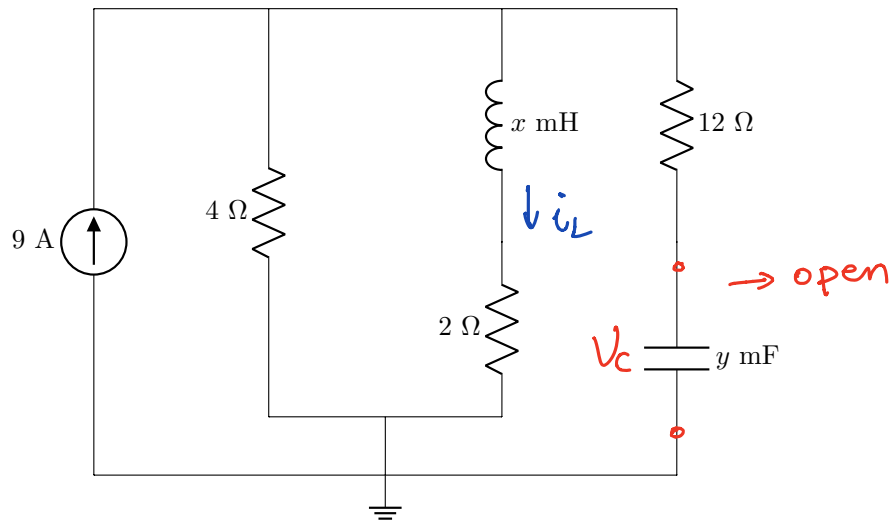
34. $i_C(t) = C \frac{dv}{dt}$, $C = 6\text{ F} \parallel 3\text{ F} = 2\text{ F}$. $v_C(t) = v_s(t)$
 $\Rightarrow i_C(t) = 2 \cdot [\alpha e^{-0.2t}]' = 2\alpha \cdot (-0.2)e^{-0.2t}$
 $= -0.4\alpha e^{-0.2t}$ A

$\Rightarrow i_C(5) = -0.4\alpha e^{-1}$ A.

35. $i_L(t) = \frac{1}{L} \int_0^5 \alpha e^{-0.2t} dt + i_L(0)$. $L = 4 \parallel 4 = 2\text{ H}$
 $= \frac{\alpha}{2} \cdot \frac{1}{-0.2} \int_0^5 e^{-0.2t} d(-0.2t) + 2$
 $= -2.5\alpha [e^{-0.2t}]_0^5 + 2$
 $= -2.5\alpha (e^{-1} - 1) + 2$
 $= 2.5\alpha - 2.5\alpha e^{-1} + 2$

Questions 36 to 37 [4]

The circuit below operates under steady-state DC conditions.
The values x and y are given in ClickUP.



36 Determine the energy stored in the inductor w_L . [2]

37 Determine the energy stored in the capacitor w_C . [2]

$$i_L = 9 \cdot \frac{4}{6} = 6A$$

$$V_C = 2 \times 6 = 12V$$

$$\textcircled{36} \Rightarrow w_L = \frac{1}{2} L i^2 = \frac{1}{2} x \cdot 10^{-3} \times 6^2 = 0.018x \text{ (J)}$$

$$\textcircled{37} w_C = \frac{1}{2} C V^2 = \frac{1}{2} y \cdot 10^{-3} \cdot 12^2 = 0.072y \text{ (J)}.$$

Question 38 [2]

Sinusoids $v_1(t)$ and $v_2(t)$ have the following general form: $A \cos(\omega t + \phi)$.

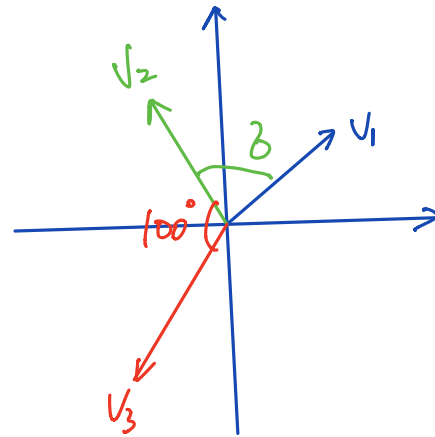
It is known that:

- v_2 leads v_1 by σ° .
- v_2 lags v_3 by 100° .

The value of σ is given in ClickUP.

38 What is the phase angle θ between v_1 and v_3 such that $0 \leq \theta \leq 180^\circ$. [2]

$$\begin{aligned} \theta + \sigma + 100 &= 360^\circ \\ \Rightarrow \theta &= 260^\circ - \sigma. \\ 0 &\leq 260^\circ - \sigma \leq 180^\circ \\ \Rightarrow 80^\circ &\leq \sigma \leq 260^\circ. \end{aligned}$$



Question 39 [2]

A current source of $i(t) = 2\cos(100t + 90^\circ)$ A is connected in parallel with a resistor of $R \Omega$, a capacitor of $C \mu\text{F}$, and an inductor of $L \text{ mH}$. Determine the phasor voltage $\mathbf{V}$ across the current source, assuming the sign of the voltage is positive. Values of R , C , and L are given in ClickUP.

39 (a) If $\mathbf{V} = A + jB$, what is the real part A ? [2]

39 (b) If $\mathbf{V} = A + jB$, what is the imaginary part B ? [2]

$$I = 2 \angle 90^\circ = j2. \quad \omega = 100.$$

$$Z_R = R$$

$$Z_L = j\omega L = j \cdot 100 \cdot L \times 10^{-3} = j0.1L$$

$$Z_C = -j \frac{1}{\omega C} = -j \frac{1}{100 \times C \times 10^{-6}} = -j \frac{10^4}{C}$$

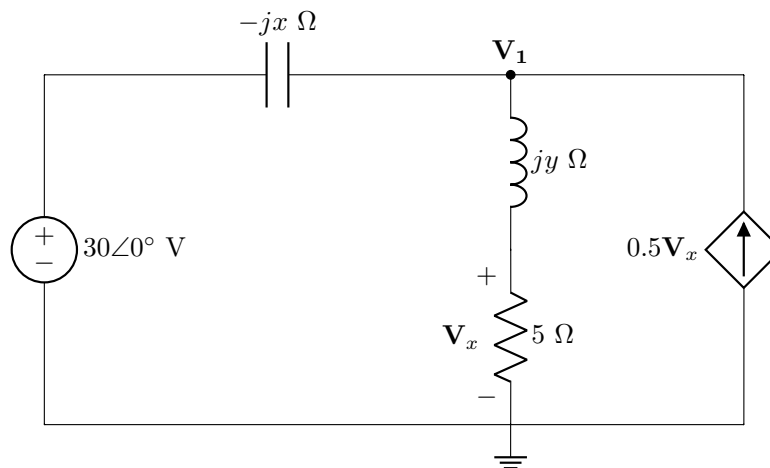
$$Z_1 = Z_L || Z_C = \frac{j0.1L \cdot -j \frac{10^4}{C}}{j0.1L - j \frac{10^4}{C}} = \frac{\frac{10^3 L}{C}}{j(0.1L - \frac{10^4}{C})}$$

$$= \frac{10^3 L}{j(0.1LC - 10^4)}$$

$$Z_2 = Z_R || Z_1 \quad (\text{use matlab})$$

$$\Rightarrow V = j2 \cdot Z_2 \quad (\text{use matlab}).$$

39

Question 40 [2]Find $\mathbf{V}_1$ in the circuit below.The values x and y are given in ClickUP.40 (a) If $\mathbf{V}_1 = C + jD$, what is the real part C ? [2]40 (a) If $\mathbf{V}_1 = C + jD$, what is the imaginary part D ? [2]

$$\frac{V_1 - 30}{-jx} + \frac{V_1}{5+jy} - 0.5 \cdot V_1 \cdot \frac{5}{5+jy} = 0$$

$$\frac{V_1 - 30}{-jx} = \frac{1.5V_1}{5+jy}$$

$$\Rightarrow -j1.5x V_1 = (5+jy)V_1 - 30(5+jy)$$

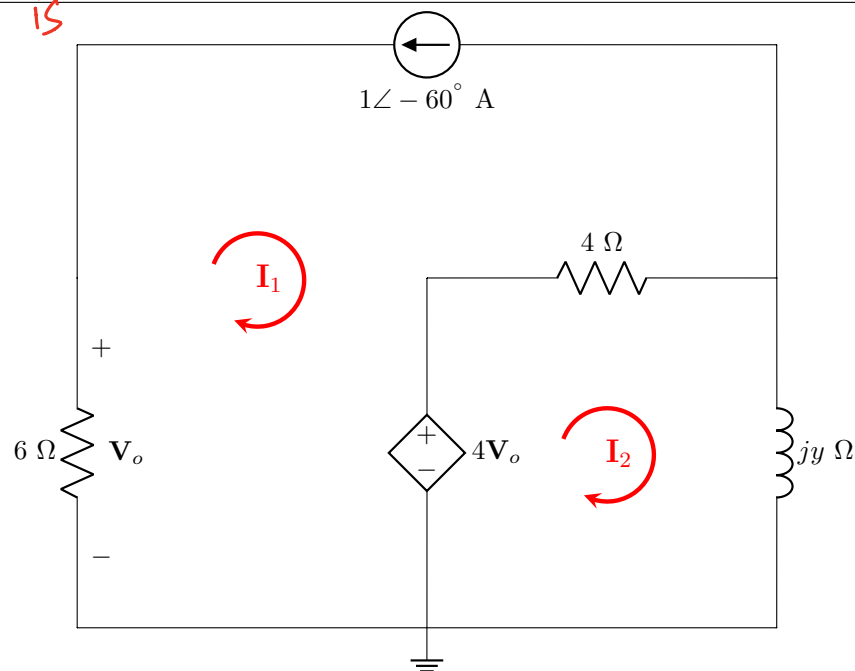
$$(5+jy-j1.5x)V_1 = 30(5+jy)$$

$$\Rightarrow V_1 = \frac{30(5+jy)}{5+j(y-1.5x)}$$

40

Question 41 [2]

In the circuit below, use **mesh** analysis to find $\mathbf{I}_2$.
The value of y is given in ClickUP.



41 (a) If $\mathbf{I}_2 = E + jF$, what is the real part E ? [2]

41 (b) If $\mathbf{I}_2 = E + jF$, what is the imaginary part F . [2]

$$\mathbf{I}_1 = -1 \angle -60^\circ, \quad V_o = 6 \angle -60^\circ$$

$$-4V_o + 4(\mathbf{I}_2 - \mathbf{I}_1) + jy\mathbf{I}_2 = 0$$

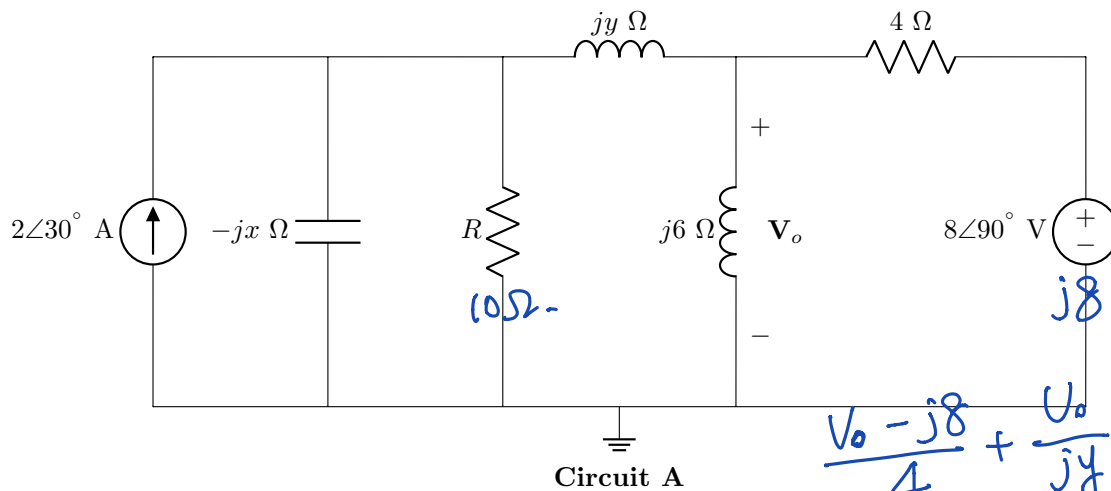
$$\Rightarrow -24 \angle -60^\circ + (4 + jy)\mathbf{I}_2 + 4 \angle -60^\circ = 0$$

$$\Rightarrow (4 + jy)\mathbf{I}_2 = 20 \angle -60^\circ$$

$$\textcircled{41} \quad \mathbf{I}_2 = \frac{20 \angle -60^\circ}{4 + jy}$$

Questions 42 to 44 [6]

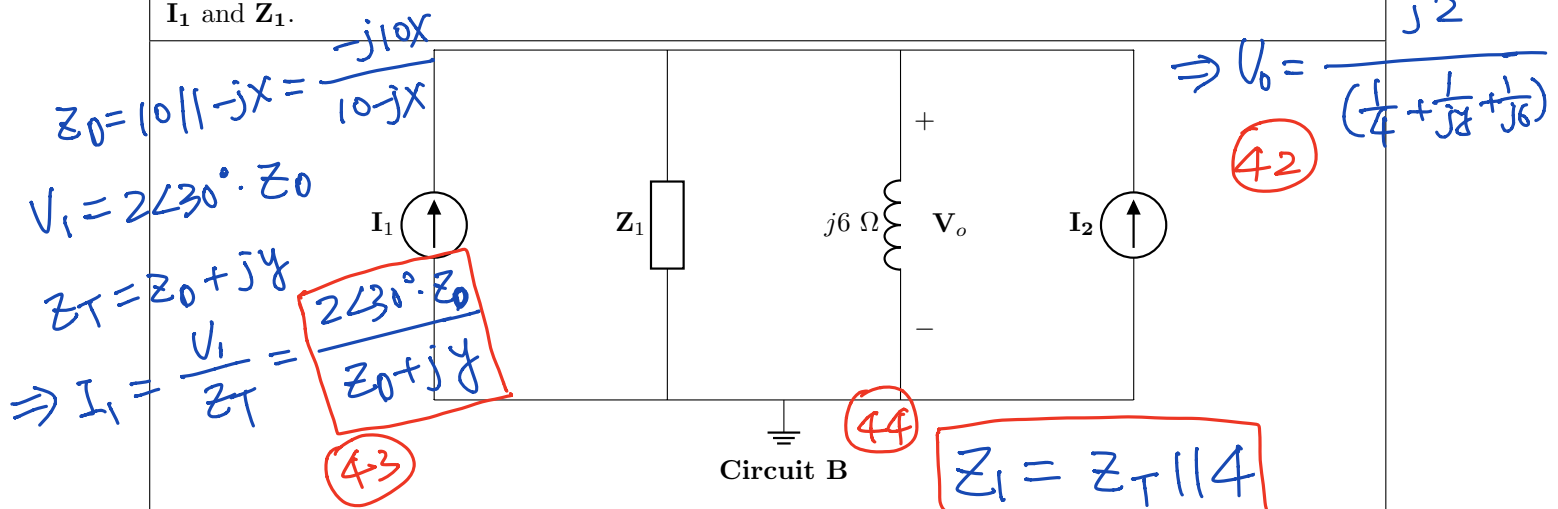
In **Circuit A**, when $R = 0 \Omega$, find the exclusive contribution, $\mathbf{V}_o^1$ from the voltage source to $\mathbf{V}_o$. The values x and y are given in ClickUP.



42 (a) If $\mathbf{V}_o^1 = G + jH$, what is the real part G ? [2]

42 (b) If $\mathbf{V}_o^1 = G + jH$, what is the imaginary part H ? [2]

When $R = 10 \Omega$, the Circuit A is transformed to **Circuit B** using source transformation. Determine $\mathbf{I}_1$ and $\mathbf{Z}_1$.



43 (a) If $\mathbf{I}_1 = K + jL$, what is the real part K ? [2]

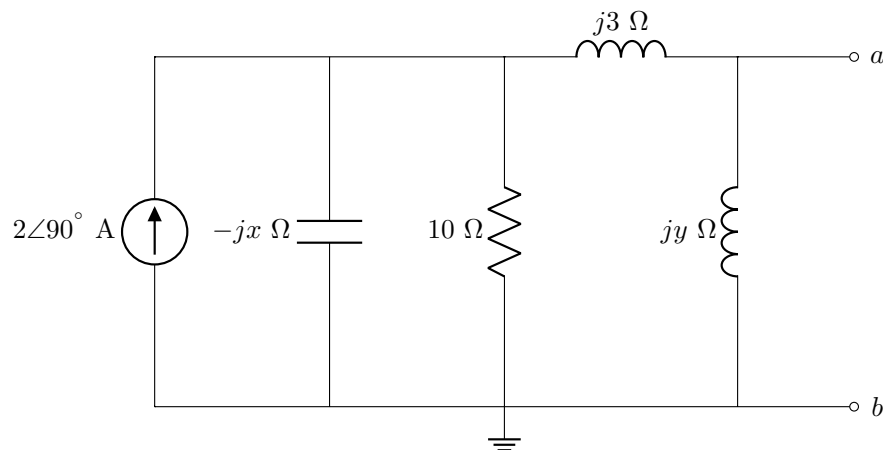
43 (b) If $\mathbf{I}_1 = K + jL$, what is the imaginary part L ? [2]

44 (a) If $\mathbf{Z}_1 = P + jQ$, what is the real part P ? [2]

44 (b) If $\mathbf{Z}_1 = P + jQ$, what is the imaginary part Q ? [2]

Questions 45 to 46 [4]

In the circuit below, find the Norton equivalent current $\mathbf{I_N}$ and impedance $\mathbf{Z_N}$ at terminals a-b. The values x and y are given in ClickUP.



- | | |
|----|---|
| 45 | (a) If $\mathbf{I_N} = S + jT$, what is the real part S ? [2] |
| 45 | (b) If $\mathbf{I_N} = S + jT$, what is the imaginary part T ? [2] |
| 46 | (a) If $\mathbf{Z_N} = U + jV$, what is the real part U ? [2] |
| 46 | (b) If $\mathbf{Z_N} = U + jV$, what is the imaginary part V ? [2] |

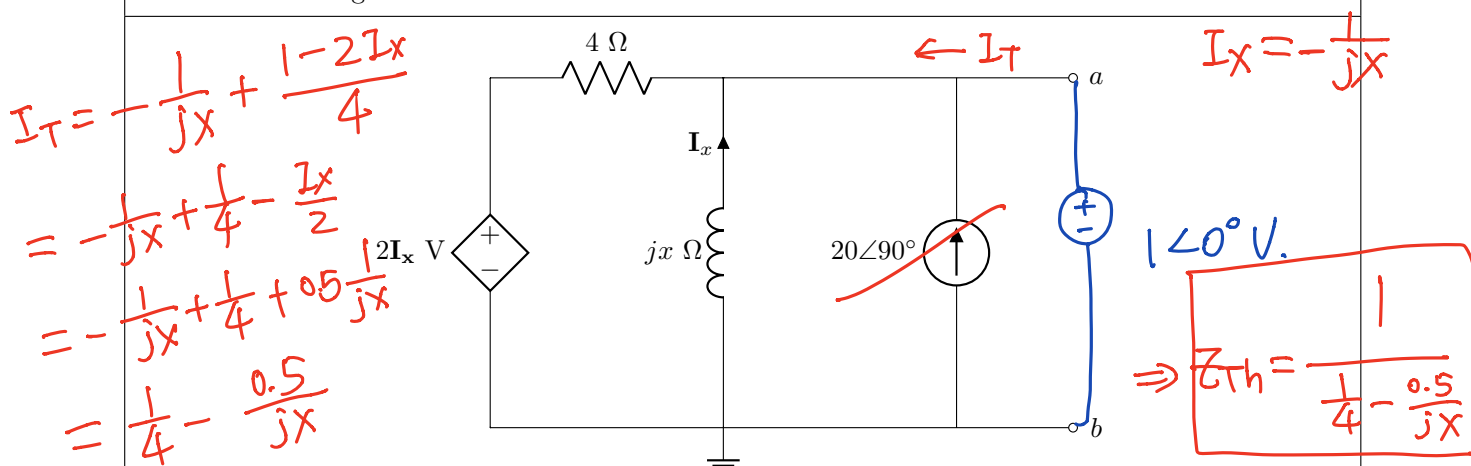
(45)
$$Z_1 = -jx \parallel 10 = \frac{-j10x}{10 - jx}$$

$$Z_2 = Z_1 + j3, \quad Z_N = jy \parallel Z_2.$$

(46)
$$I_N = j2 \cdot \frac{Z_1}{Z_1 + j3}.$$

Questions 47 to 48 [4]

In the circuit below, find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals a-b. The value of x is given in ClickUP.



- | | |
|----|---|
| 47 | (a) If $V_{Th} = X_1 + jY_1$, what is the real part X_1 ? [2] |
| 47 | (b) If $V_{Th} = X_1 + jY_1$, what is the imaginary part Y_1 ? [2] |
| 48 | (a) If $Z_{Th} = X_2 + jY_2$, what is the real part X_2 ? [2] |
| 48 | (b) If $Z_{Th} = X_2 + jY_2$, what is the imaginary part Y_2 ? [2] |

$$\frac{V_{Th} - 2I_x}{4} + \frac{V_{Th}}{jx} - j20 = 0$$

$$\Rightarrow \frac{V_{Th} - 2 \cdot (-\frac{V_{Th}}{jx})}{4} + \frac{V_{Th}}{jx} = j20$$

$$\frac{V_{Th}}{4} + 0.5 \frac{V_{Th}}{jx} + \frac{V_{Th}}{jx} = j20$$

$$(\frac{1}{4} + \frac{1.5}{jx}) V_{Th} = j20 \Rightarrow V_{Th} = \frac{j20}{\frac{1}{4} + \frac{1.5}{jx}}$$

(47)

Total Section 3 / Totaal Afdeling 3: [32]

Grand Total: [96]